Backbone feature-map evolution of the fine-tuned YOLOv5 detector on an IR laser soil image

Input 640×640

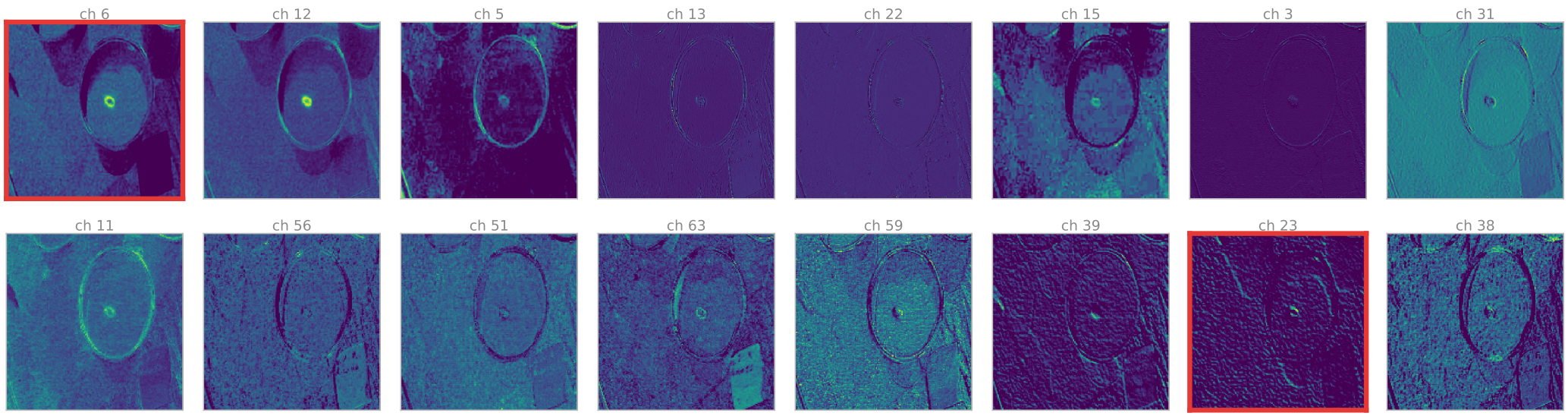


Each row: the 8 channels responding most strongly inside the detected laser region.
Early layers (P1, P2) keep full spatial detail; generic edges, contrast and color cues fire everywhere.
Middle layers (P3, P4) trade resolution for depth; activations organize around the laser spot
core, halo and boundary. Deep layers (P5, SPPF) are coarse but semantic; nearly all remaining
activation sits on the object region and background clutter is discarded.
Red outline: most laser-selective channel (highest in-box vs. global activation ratio).

P1/2

Conv · 320×320×32

edges · contrast ·
color transitions



P2/4

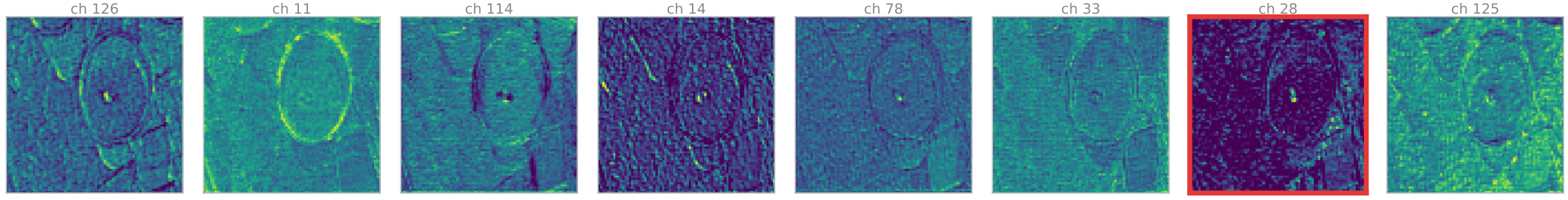
C3 · 160×160×64

fine textures ·
local primitives

P3/8

C3 · 80×80×128

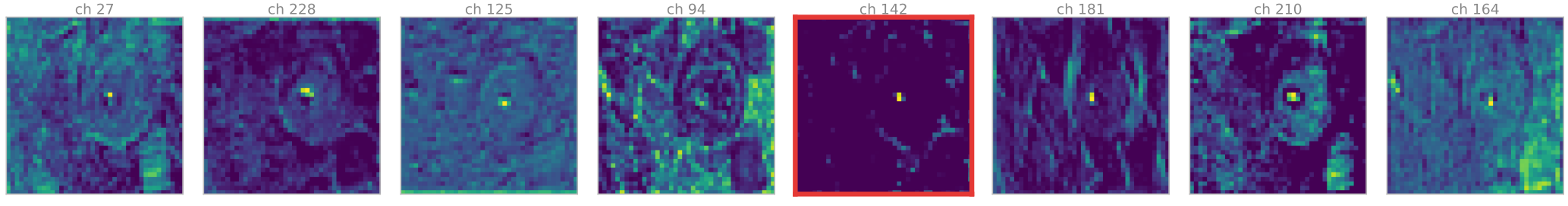
spot components:
core · halo · rim



P4/16

C3 · 40×40×256

part assemblies ·
object contours



P5/32

SPPF · 20×20×512

whole-object semantics ·
background suppressed

